

ATAI 8WD Gen-1 Safe Cockpit — 100

Engineering Documents

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1. Program Definition & System Architecture

ENG-001 — Gen-1 Safe Cockpit Program Charter
ENG-002 — Top-Level System Requirements Specification
ENG-003 — System Architecture Definition Document
ENG-004 — Product Configuration Baseline
ENG-005 — Program Assumptions and Constraints Register
ENG-006 — Stakeholder Requirements and Priority Matrix
ENG-007 — System Safety Objectives and Design Targets
ENG-008 — Technical Risk Register and Mitigation Plan
ENG-009 — Engineering Interface Control Overview
ENG-010 — Development Roadmap and Maturity Milestones

2. Mission, Use Cases & Operating Envelope

ENG-011 — Mission Profile Definition Document
ENG-012 — Primary Use Case and Scenario Catalogue
ENG-013 — Off-Nominal Use Case and Contingency Catalogue
ENG-014 — Operating Envelope Definition and Boundaries
ENG-015 — Environmental Conditions Specification
ENG-016 — Payload, Crew, and Occupancy Assumptions
ENG-017 — Range, Endurance, and Duty Cycle Requirements
ENG-018 — Performance Mission Allocation Summary
ENG-019 — Operational Concept of Use Document
ENG-020 — Dispatch, Storage, and Ground Handling Requirements

3. Structural Design & Materials

ENG-021 — Primary Load Path Definition
ENG-022 — Structural Design Criteria and Margin Policy
ENG-023 — Airframe Geometry and Packaging Definition
ENG-024 — Cabin Pressure Shell Structural Requirements
ENG-025 — Frame, Bulkhead, and Hardpoint Specification
ENG-026 — Materials Selection and Qualification Basis
ENG-027 — Composite Laminate Definition and Allowables
ENG-028 — Metallic Component Specification and Corrosion Control
ENG-029 — Structural Fatigue and Damage Tolerance Plan
ENG-030 — Structural Integrity and Proof Load Requirements

4. Propulsion, Lift Fans & Powertrain

ENG-031 — Propulsion System Requirements Specification
ENG-032 — Lift Fan Architecture Definition
ENG-033 — Thrust Allocation and Control Strategy
ENG-034 — Motor and Drive Unit Specification
ENG-035 — Propulsor Integration and Installation Requirements
ENG-036 — Rotor, Duct, and Inlet Flow Requirements
ENG-037 — Powertrain Thermal Management Specification
ENG-038 — Propulsion Performance Map and Operating Limits
ENG-039 — Propulsion Fault Detection and Protection Logic
ENG-040 — Propulsion Ground Test Requirements

5. Energy System & Electrical Architecture

ENG-041 — Energy Storage System Requirements
ENG-042 — Battery Pack Architecture Definition
ENG-043 — High-Voltage Electrical Architecture Specification
ENG-044 — Power Distribution and Load Shedding Logic
ENG-045 — Electrical Protection and Isolation Requirements
ENG-046 — Charging Interface and Ground Support Requirements
ENG-047 — Cable Routing, Harness, and Connector Standard
ENG-048 — Energy Monitoring and State Estimation Specification

ENG-049 — Thermal-Electrical Safety and Containment Requirements

ENG-050 — Electrical Power Quality and Transient Requirements

6. Flight Control, Avionics & Software

ENG-051 — Flight Control System Requirements Specification

ENG-052 — Control Laws Architecture and Allocation Document

ENG-053 — Avionics Architecture Definition

ENG-054 — Navigation and Position Estimation Requirements

ENG-055 — Sensor Suite Specification and Performance Budget

ENG-056 — Flight Computer Hardware Requirements

ENG-057 — Software Development Assurance Plan

ENG-058 — Real-Time Operating Concept and Timing Budget

ENG-059 — Command, Monitoring, and Health Management Logic

ENG-060 — Data Logging, Telemetry, and Diagnostics Specification

7. Safety, Escape & Survivability

ENG-061 — System Safety Assessment Plan

ENG-062 — Hazard Analysis and Risk Classification Register

ENG-063 — Functional Safety Requirements Specification

ENG-064 — Emergency Shutdown and Safe State Logic

ENG-065 — Escape System Architecture Definition

ENG-066 — Emergency Egress and Access Requirements

ENG-067 — Fire Detection, Suppression, and Containment Specification

ENG-068 — Crashworthiness and Occupant Protection Requirements

ENG-069 — Survivability and Post-Landing Safety Assessment

ENG-070 — Emergency Response and Rescue Interface Document

8. Human Factors, Cabin & Life Support

ENG-071 — Crew Interface Requirements Specification

ENG-072 — Cockpit Layout and Reach Envelope Definition

ENG-073 — Visibility, Display, and Alerting Requirements

ENG-074 — Seating, Restraint, and Posture Accommodation Specification

ENG-075 — Cabin Environmental Control Requirements

ENG-076 — Oxygen, Pressurization, and Breathing Support Specification
ENG-077 — Acoustic, Vibration, and Comfort Criteria
ENG-078 — Human Machine Interface and Workload Assessment
ENG-079 — Crew Procedures and Checklists Definition
ENG-080 — Emergency Human Factors and Escape Usability Review

9. Manufacturing, Assembly & Maintainability

ENG-081 — Manufacturing Process Architecture
ENG-082 — Assembly Sequence and Build Philosophy
ENG-083 — Tooling, Jigging, and Fixture Requirements
ENG-084 — Production Quality Assurance Plan
ENG-085 — Configuration Management and Serialization Plan
ENG-086 — Inspection and Acceptance Criteria Document
ENG-087 — Maintainability and Line Replaceable Unit Strategy
ENG-088 — Access Panel and Serviceability Requirements
ENG-089 — Spare Parts, Repair, and Depot Support Plan
ENG-090 — Production Readiness and Industrialization Checklist

10. Verification, Validation, Certification & Test

ENG-091 — Verification and Validation Master Plan
ENG-092 — Requirements Traceability Matrix
ENG-093 — Test Strategy and Test Pyramid Definition
ENG-094 — Simulation and Digital Twin Validation Plan
ENG-095 — Ground Test Campaign Requirements
ENG-096 — Component Qualification Test Specification
ENG-097 — System Integration Test Requirements
ENG-098 — Flight Test Plan and Envelope Expansion Strategy
ENG-099 — Certification Basis and Compliance Mapping
ENG-100 — Final Acceptance, Conformity, and Release Criteria

This document tree defines the initial engineering knowledge architecture for the ATAI 8WD Gen-1 Safe Cockpit program.